

2.4 General safety management

2.4.1 Safety management systems

Rail operators must obtain a **safety certificate** for operating trains, or a **safety authorisation** for infrastructure managers (which includes station managers).

The status of a safety certificate and a safety authorisation is the same, in that they allow access to the railway; however, their scope will be different. Safety certificates and safety authorisations have a maximum period of validity of five years.

There are various exemptions for the requirement to have either a safety certificate or authorisation. Some of these are shown below.

- Those operating only in engineering possessions.
- Those operating on non-mainline transport systems where the maximum permitted speed is 40 kph or less.
- Tramways.
- Railways in factories and most harbours.
- Railways within maintenance or goods depots or military establishments.

In order to obtain a safety certificate or safety authorisation the operator must implement a safety management system that includes the following.

- A statement of the safety policy.
- Targets for the maintenance and enhancement of safety, and plans and procedures for reaching those targets.
- Procedures to meet relevant technical and operational standards or other requirements set out in national safety rules, for example:
 - procedures and methods for carrying out risk assessments and implementing risk control measures
 - provisions for employee training and maintenance of competency
 - arrangements for the provision of information relevant to safety
 - procedures and formats for the documentation of safety information
 - procedures to control the layout of and changes to vital safety information
 - procedures to ensure that accidents, incidents, near misses and other dangerous occurrences are reported, investigated and analysed and that necessary preventative measures are taken
 - provision of plans for action, alerts and information in the case of an emergency
 - provisions for internal auditing of the safety management system.

In practice, a train operating company, as a vehicle operator, or Network Rail, as an infrastructure manager, will stipulate safety conditions in accordance with their safety management systems that need to be complied with for a particular contract. These will be incorporated into the licensing arrangements (*refer to 2.3*) and the construction phase plan for the project.

However, if contractors operate trains or carry out infrastructure maintenance outside of possessions then the requirements for the safety certificate, authorisation and safety management system will apply.

2.4.2 Sentinel scheme

If work on a railway entails working on or near the line, workers must have completed appropriate track safety training, including industry common induction (ICI) and a personal track safety (PTS) course, and be issued with, and be in possession of, a sentinel smartcard. This smartcard must be presented for authentication each time the worker goes on or near the line.

Card authentication can be achieved by a number of means including swiping the card with a smartcard reader, iPhone or Android check, QR code scanning, web check and interactive voice recognition (IVR).

Before undergoing training for personal track safety, workers must be certified medically fit for work in a railway environment, and undergo screening for drugs and alcohol. The standards of medical fitness are laid down in Railway Group and Network Rail Standards.

Any workers who do not meet the full medical standards may still be permitted to work on the railway subject to risk assessment and work restrictions being applied.



Workers who successfully undergo PTS training receive a *Keypoints – Personal track safety (PTS)* health and safety booklet outlining safe working procedures and their responsibilities when working on the railways. A PDF copy of the booklet is available online.

2.4.3 Competency and training in track safety supervision

The Office of Rail and Road (ORR), having taken over responsibility for health and safety on the railways (from the Health and Safety Executive (HSE) in 2006), introduced a competence management system (CMS), the structure of which is explained in the ORR publication *Developing and maintaining staff competence (Railway safety publication 1)*.

The publication is a guide for people who are responsible for ensuring and maintaining the competency of operatives whose work could have an impact on the rail network. Managers responsible for maintaining the CMS should be familiar with the detail within the ORR publication.

2.4.4 Personal track safety card

This competency allows qualified workers to walk on or near the line alone, to access or egress the site only, without the requirement to have either a COSS or SWL present. PTS card carriers at the most basic level cannot carry out **any** work on or near to a line. The PTS cards are also endorsed with more detailed information specific to working on tracks that use conductor rail equipment (CRE) and overhead line equipment (OLE). The person in charge will have to ensure that operatives' cards are endorsed appropriately before allowing them near the tracks.

The competency card allows qualified workers to work on alternating current (AC)/direct current (DC) line areas under the control of a COSS or SWL, providing the staff hold AC/DCCR competence and PTS on the sentinel database.

This competency is valid for two years. Upon successful completion of a new course and any additional modules, the online sentinel database will be updated. However, another card will not be issued. The new sentinel smartcards replace the previous cards, which are no longer valid for access to Network Rail infrastructure.

2.4.5 Lookout or site warden

This competency allows a qualified person to have the following.

- Lookout duties to provide enough warning to allow everyone involved in the work to reach a position of safety at least ten seconds before any train or vehicle arrives at the site of work (formerly known as the *red zone*).
- Site warden duties to maintain the safe limits of a protected area (formerly known as the *separated green zone*).

2.4.6 Person in charge of possession

This competency enables a person to carry out person in charge of possession (PICOP). This role is specific to Network Rail. To be a PICOP (or engineering supervisor), the person's details demonstrating COSS experience within the last 24 months must be held in the employer's competence management system. The PICOP will also be required to meet in full certain experience criteria. PICOP competence itself does not qualify anyone to do COSS work, which requires a separate COSS qualification.



For information on developing and maintaining staff competence refer to Appendix C.

2.4.7 Safety critical work

The ROGS Regulations lay down a procedure to ensure that those engaged on a transport system in a work situation do so in a manner that will not affect the health and safety of people using that transport system. All workers must be competent and physically fit to carry out their allotted tasks. They must also work in such a way that the number of hours worked, combined with the rest periods taken between work, does not increase the risk of fatigue to operatives.

It is obvious that any work carried out on the railway (whether to the track, the signalling system or the trains themselves) could have disastrous effects if it is not carried out correctly. The whole reasoning behind the ROGS Regulations is to ensure that only healthy, physically fit and competent persons carry out work on the railways. These regulations should ensure that the work will be carried out to such a high standard that the risk of disaster will be negligible.

2.4.8 Working alone

People must never work alone unless they are qualified to carry out the duties of the:

- controller of site safety (COSS)
- individual working alone (IWA)
- safe work leader (SWL).

In addition, workers must have permission from and be in communication with the:

- protection controller (PC)
- engineering supervisor (ES).

2.4.8.1 Going on or near the line alone

In most instances individuals should never work alone on or near trackside. The only time when they can be alone is when they are walking and only if they comply with these requirements.

- Have a valid certificate of competence in personal track safety issued by the employer.
- Wear the correct high-visibility clothing issued by the employer.
- Have a suitable lamp if they will be on or near the line during darkness, poor visibility or if they will enter a tunnel.



The only occasion when a worker can go on or near the line without a COSS or SWL is when they are walking alone.

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2.4.9 Train movements



A train travelling at 80 mph will cover nearly 36 m (120 feet) per second.

If it takes 10 seconds to realise that a train is approaching and move off the track to a place of safety, in that time the train will have covered nearly 360 m (1,200 feet).

When anyone is working on or near a railway line, the driver of an approaching train should blow the train whistle or horn to warn of the train's approach, although this must not be relied upon.

The worker, in return, must acknowledge that they have heard the whistle, and have moved to a position of safety, by raising one arm straight in the air above their head.

All work on or near the line must be pre-planned. If lookout protection is to be utilised, then the COSS or SWL needs to assess the position and number of lookouts to ensure that warnings are provided.

The time required is decided by the COSS or SWL by conducting a risk assessment, and will depend on the speed of the approaching train and the length of time necessary to remove staff and place tools away from the running lines to enable the whole group to be standing in a position of safety for at least 10 seconds before the train passes the worksite.

The speed of the train is always taken as the maximum permissible speed at that location.

When setting up a safe system of work, a COSS or SWL needs to determine the distance at which they must be able to see a train approaching. The *Rule book* (GE/RT8000) defines safety distances according to the following factors.

If the **maximum speed** is:

- 100 mph or less, a person is in a position of safety if they are at least 1.25 m (4 feet) from the nearest line on which a train can approach;
- over 100 mph, the distance increases to 2 m (6 feet 6 inches).



A train sighting chart and an explanation of how it is used is included at Appendix A.

2.4.10 Communications protocol

You must make sure you properly understand the meaning of all messages, whether they are communicated by phone, radio or face-to-face.

- Make sure you are talking to the right person.
- Give your location, if using a phone or radio.
- Give your name and the name of your employer.
- State what task you are performing.
- If necessary, let the person know how you can be contacted.
- Use the phonetic alphabet to make sure names and locations that are difficult to pronounce are fully understood.
- Never use the words 'not clear' to describe a line that is obstructed; always use 'line blocked'.
- You must say numbers one at a time (for example, you should say 8107 as 'eight, one, zero, seven'). There are exceptions to this (such as when giving the time or when referring to a rule book module or handbook).

If you are receiving a message, make sure you fully understand it. You must repeat the message back so that the other person knows you correctly understand it.

Ensure the following to help make sure your message is fully understood when using a telephone or radio.

- Speak with the mouthpiece close to your mouth and speak directly into the mouthpiece.
- Talk slightly slower than normal using a natural rhythm.
- Use your normal level of volume when speaking.
- Avoid using hesitation sounds (for example 'um' and 'er').
- Use clear sentences.
- Get the person to repeat your message back to you.

During any conversation, one person must always take **lead responsibility**.

The person who must take lead responsibility depends on the task being carried out.

Examples are shown in the table on the next page.

Lead responsibility	When communicating with
Electrical control operator (ECO)	Anyone
Signaller	Anyone except the ECO
PICOP (person in charge of the possession)	Anyone except the ECO or signaller
Engineering supervisor	Each COSS/PC
Route-setting agent	Points operator
Protection controller (PC)	Each COSS
COSS	Members of the work group
COSS	Each lookout/site warden

If it is not clear who has lead responsibility, or if two people carrying out the same task are communicating with each other, the person who begins the conversation must always take lead responsibility.

Phrases to use when using a radio or telephone	
Phrase	Meaning
This is an emergency call	This message provides information that needs immediate action to prevent death, serious injury or damage
Repeat back	Repeat all of the message back to me
Correction	I have made a mistake and will now correct the word or phrase just said

Other phrases to use when using a radio and only one person can be heard at a time	
Phrase	Meaning
Over	I have finished my message and am expecting a reply
Out	I have finished my message, no reply is expected

The **phonetic alphabet** must be used:

- to identify letters of the alphabet
- to spell words and names that are hard to say, or misunderstood
- if there is interference on the radio or phone
- when quoting the identity of signals or points
- when quoting train descriptions.

The phonetic alphabet

A – Alpha	H – Hotel	O – Oscar	V – Victor
B – Bravo	I – India	P – Papa	W – Whisky
C – Charlie	J – Juliet	Q – Quebec	X – X-ray
D – Delta	K – Kilo	R – Romeo	Y – Yankee
E – Echo	L – Lima	S – Sierra	Z – Zulu
F – Foxtrot	M – Mike	T – Tango	
G – Golf	N – November	U – Uniform	

2.4.11 Confined spaces

There are work areas on a railway that may be classified as confined spaces. These include box girders, closed arches, culverts, station basements and inspection pits. Having identified a confined space, the following questions should be asked.

- Is there any way of carrying out the work task without entering that space?
- If there is a need to enter that area, how long is it going to take for the work to be completed?

Entry into confined spaces must be in accordance with the Confined Spaces Regulations 1997 and controlled by a permit-to-work system.



For further information on working in a confined space, refer to Chapter D08 Confined spaces.

2.4.12 Lineside fencing

The boundary of railway property is protected by the erection of lineside fencing. Any work on this fencing must be carried out in a controlled way to prevent inadvertent or unauthorised access to the railway.

Some sections of fencing, due to the close proximity of electrified lines, should be connected together by bonding to earth. This will avoid the risk of electric shock to people coming into contact with the fence.

Renewal of fencing should be carried out in such a manner that the access created can be controlled by use of personnel or temporary fencing during these periods.



Whenever the site has to be left unattended, the fencing must be left in such a state that access cannot be gained.

2.4.13 Lineside vegetation

Removal of lineside vegetation can be hazardous for those involved. Some hazards are shown below.

- The close proximity of electrified lines.
- Fire.
- The types of tool used.
- Discarded rubbish.
- Injuries from sharps (such as discarded needles or broken glass).
- Vermin infestation.
- Injurious plants (such as giant hogweed).
- Insect stings (such as bees or wasps).

Removal of vegetation must be carried out in such a way that it does not come within 2.75 m of overhead electrified power lines and does not cause an obstruction of the track. Consideration also needs to be given to the presence of sub-surface cables, as many of these have previously been cut during the removal of vegetation.

Vegetation must not be burnt, because of the risk of smoke obstructing a train driver's vision.

Tools used for vegetation clearance can present hazards of their own. Safe systems of work should include the setting up of exclusion zones around hazardous operations, where tools such as chainsaws and flails are being used. To control the risk of hand-arm vibration syndrome (HAVS) it is important that you understand trigger times, use of job rotation, taking regular breaks and reporting any symptoms to your manager immediately.

Rubbish attracts vermin, which in turn can create the hazard of Weil's disease. There is also the hazard of discarded hypodermic syringes. These should be removed while taking every precaution against wounds from contaminated sharps. If you come into contact with an unprotected syringe, you should seek immediate medical advice.



For further information on HAVS refer to Chapter B12 Vibration.

2.4.14 Working on stations and platforms

Although passengers use station platforms regularly and feel safe doing so, when work is being carried out a platform can be just as dangerous as the open track. Work on platforms and the station infrastructure can be in various guises, such as painting, billboard erection, maintenance work and installation work.

Whatever the work being carried out, the inherent risks are the same.

- Passage of trains.
- Presence of electrified tracks and overhead cables.
- Movement of station staff.
- Movement of passengers and their luggage.
- Risk of fire.

Work on stations and platforms should be planned in the same way as work on any other part of the railway system. This will ensure the health and safety of all concerned. All work will require approval by the station manager.

Consideration must be given to the following points whenever work is undertaken on stations or their platforms.

- Work must not be allowed to interfere with the normal operation of trains.
- Work must not be allowed to interfere with the normal operation of the station.
- Work must be carried out safely, bearing in mind the presence of electrified tracks and overhead cables.
- The use of flammable substances may create additional hazards.
- The way in which materials to be used on site will be transported and stored.

Over the years, work being carried out on stations and their platforms has resulted in several fatal accidents. This has been due to the lack of proper planning, failure to isolate power supplies and lack of proper control.

A safety-conscious approach is now being adopted in the way that contracts are drawn up for tendering. It involves bringing into the contract the requirements of the Construction (Design and Management) Regulations 2015, the ROGS Regulations and the various Railway Group Standards that govern the way in which people work in a railway environment.

Below are the golden rules to remember when working on station platforms.

- Beware of passing trains.
- Beware of electrified tracks and overhead cables.
- Do not obstruct a train driver's vision of signalling equipment.
- Do not obstruct passenger routes.
- Know the fire and other emergency procedures.

Working on platforms within four feet of the platform edge, when carrying out engineering or technical work, is classed as working on or near the line. People working there are required to hold a valid personal track safety certificate.

A system of creating a 'high street environment' can be accomplished if the work is not on or near the line.

This can be done by erecting a physical barrier or by appointing a site keeper who is responsible for preventing the work encroaching on the railway infrastructure.

If you are creating a high street environment within the station, this system must be agreed with the station manager to ensure that there is no conflict with the travelling public or with station operations. You may still need a safe work pack (SWP) to enable you to access the area known as 'on or near the line' in order to access your high street environment working area.

2.4.15 Work sites

Work sites are categorised depending upon the level of protection required.

2.4.16 Possession of the line

When major engineering work is to be undertaken, a possession will be required.

This is an agreement with the signaller to give over control of the line, but control must be returned to the signaller when requested.

Possession of the line is taken by the person in charge of possession (PICOP) from the signaller, and the PICOP is the only one who can allow trains in or out of the line(s) under possession. This procedure must be planned well in advance of the possession being required, otherwise it will not be granted.

Extra care is needed if one line is under possession and the line next to it is still open to train movements. Steps have to be taken to prevent people straying from the line under possession onto the open line (*refer to Figure 1 in Appendix B*).

2.4.17 Blockages of the line

Under this arrangement, trains are not permitted to run. The line block is established by a COSS or SWL in order to protect staff from train movements but not traction current. Signals are set to danger in advance of the works.

This arrangement may require additional protection (for example, in the form of placing detonators (railway fog signals) on the line on the approach to the work site).

Extra care is needed if one line is blocked and the line next to it is still open to train movements. Steps have to be taken to prevent people straying from the line that is blocked onto the open line.

2.4.18 Considerations when undertaking track work

2.4.18.1 Fencing and separation

To prevent people at a work site straying onto lines that are open to train movements, physical reminders can take the form of blue, 1 m high fencing, or black and yellow tape extending beyond the length of the work site. The fencing must be supported to prevent it being dragged under passing trains. An alternative type of fence is a magnetic rigid barrier.

Fencing acts as a continual reminder to people on site that, beyond the fencing, the track is open to movement of trains. Therefore, a safe system of work exists, providing no-one strays beyond the fencing (*refer to Figures 2 and 3 in Appendix B*).

An alternative to fencing is to use site wardens to prevent people straying onto lines that are open to train movements.

The wardens are positioned by the COSS at least 2 m away from the nearest rail of any line on which a train may run, to warn anyone who moves towards that 2 m point (*refer to Figure 4 in Appendix B*).

Employing this system as a means of warning depends upon the alertness of the site wardens ensuring their colleagues do not stray beyond the limits set by the COSS, within 2 m of the open line.

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2.4.18.2 Prohibition of work

At some locations, as detailed in Network Rail's hazard directory, it is not permitted to carry out works with trains running (formerly known as the *red zone*). An alternative safe system of work must be established at these locations.

2.4.18.3 Work planning

All work on or near the line must be planned by a safe system of work planner (SSOWP). The planner provides a safe system of work pack, which is issued to the COSS at least one shift prior to the planned work item. The COSS must verify the proposed SSOW with the planner in advance of the work. Then the COSS must implement the SSOW, as planned. Any changes thereafter must be formally agreed by a responsible manager before work can be allowed to continue. SSOWP is a sentinel competence.

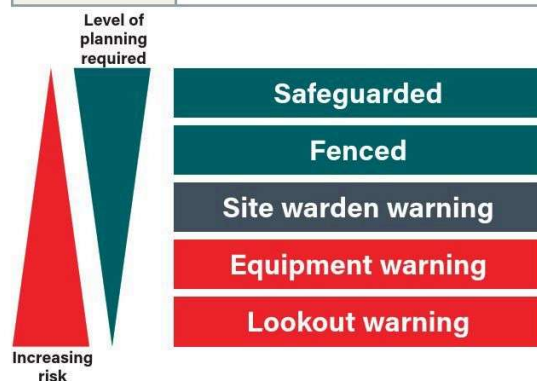
The protection arrangements chosen must be the best available from the following list.

Best	Safeguarded.
2nd	Fenced.
3rd	Separated.
4th	Equipment warning – Permanent.

Together, these will lead to work that is better planned and co-ordinated to provide a higher level of safety. They will also help to drive further improvements through the introduction of systems that offer better protection and are simpler to use.

These include track circuit operating devices (T-CODs), automatic track warning systems (ATWS) and plans for a simplified version of protection arrangements to make it easier to obtain the separation of people from train movements.

5th	Equipment warning – Human activated.
6th	Equipment warning – Portable.
Last resort	Lookout warning.



2.5 Plant and equipment

2.5.1 Mobile plant

Any mobile plant used on the railways must not create a danger to the passage of trains or fixed rail infrastructure. This entails the COSS setting up a safe system of work for the use of that plant within the designated work area.

A work package plan and task briefing sheets should identify how plant will be brought to site, used on site and taken off site.

Some mobile plant is capable of being driven on both the road and on rails (*refer to illustration on right*) and such plant must conform to the appropriate Railway Group Standard. Contractors using plant that can move on rails must be a plant operating scheme (POS) provider.

The method of getting the plant to and from site must be carefully planned. Contractors should ensure that the plant does not damage other vital equipment (such as structures, signalling equipment (including cables) and communications equipment).

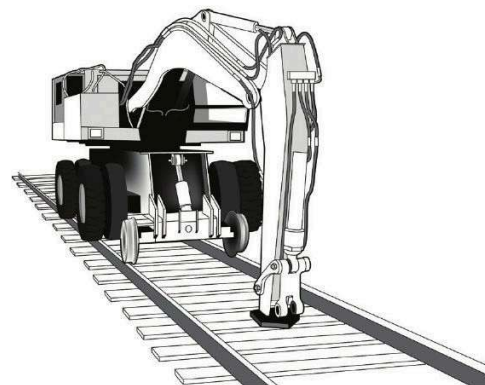
Ways of protecting the various items of equipment must be clearly outlined in the work package plan and task briefing sheets for the work. When working on site with mobile plant, various factors have to be taken into consideration, such as the following.

- The presence of electrified lines.
- The presence of underground services.
- The presence of signalling equipment.
- Working on embankments.
- Obstructing train drivers' vision and sighting of signals.
- The use of lights on road vehicles.
- The distance from running lines open to traffic (any line open (ALO)).

These are in addition to the normal precautions that have to be taken by plant operators in any working environment.

Where items of plant are used in relation to railway infrastructure, they must be used and controlled only by competent persons.

Certificates of competence to operate items of plant must be carried, along with means of identification. These credentials can be asked for at any time. Failure to provide them when requested will result in the person being removed from site.



Plant and machinery must conform to the appropriate Railway Group Standard

2.5.2 Cranes and mobile lifting equipment

Tower cranes, mobile cranes and other mobile lifting equipment present risks to railways during rigging, use, maintenance and de-rigging. The risk of a crane collapsing or failure of loads during lifting operations could lead to fatalities; damage to the tracks, power lines or rolling stock; or injury to passengers and/or railway staff.



In recognition of these serious risks the Construction Plant-hire Association and Network Rail have developed guidance for tower cranes and mobile cranes alongside railways, which are free to download from the CPA website.

The guidance provides detailed information on early planning and liaison with the rail authorities, site set up and foundation requirements for rigging and de-rigging of cranes and the lines of communication through planning, operation and requirements for supervision, maintenance and thorough examination of cranes.

In essence, the guidance is a guide to the good practice that must be followed to protect people and property from harm or damage.



For general information on lifting operations and lifting equipment refer to Chapter C07 and Chapter C08.

2.5.3 Scaffolding and mobile access towers

Scaffolding work in a railway environment is a specialist activity, requiring the scaffolders to be competent not only in scaffolding but also in railway work.

A train requires clearance between itself and any structure that it passes. These clearances vary according to the train speed, the type of rolling stock, the presence or non-presence of curves and the cant of the track.

The required clearances need to be observed when erecting a scaffold or mobile access tower to ensure that it cannot be struck by passing trains, and that the movement of air caused by trains passing does not affect the structure.

A thorough survey of the site must be made prior to the erection of any scaffold or mobile access tower. The survey must consider the needs of railway personnel who need an unobstructed view of equipment in order to carry out their work safely.

Failure to carry out this survey correctly could have serious consequences.

Erection of scaffolds and mobile access towers on the railway must consider the following factors.

- Movement of trains.
- Movement of station staff and passengers.
- Proximity of electrified lines.
- Location of signals and other equipment used in rail operations.



It is vital that scaffolds or mobile access towers are not erected in positions that will obstruct signals and other equipment that need to be seen by train drivers.

Particular standards exist and must be complied with for the erection and use of scaffolds or mobile access towers, adjacent to or over the tracks, and these must be carried out in such a way that train movements are not affected.

When scaffolding has to be erected where electrified lines are situated, the method of erection, working and dismantling must take account of the hazards presented by the presence of the electrified lines.

Some sections of scaffolding, due to the proximity of electrified lines, should be bonded to earth to avoid the risk of electric shock to people coming into contact with the scaffold. The erection or dismantling of a scaffold or mobile access tower must not commence until an isolation permit has been received to indicate that it is safe to work near the electrified lines.

The need for the isolation of electrified lines during scaffold erection and dismantling is of paramount importance to avoid contact with live lines when moving scaffold poles around.

The stoppage of trains during this process should also be considered, because of the possibility of the incomplete scaffold collapsing onto the track.



Even if an electrical isolation permit has been received, it will not prevent the movement of non-electrified trains or runaway rolling stock.

Signalling systems operate when the wheels and axles of a train running along the rails create an electrical connection of one rail to the other. This allows a small current to flow, which completes the track circuit to operate signal lights, indicating that the train is in that section.

Extreme care must be taken when erecting or dismantling a scaffold or mobile access tower over tracks. If scaffold tubes fall, or are placed across rails, the signals may change. This will result in false signals being given to train drivers (for example, green lights instead of red or yellow ones), which could have serious results.

2.6 Electrified lines

A large amount of the railway network is electrified, creating a hazard to anyone working in the vicinity of these sections. There are three different types of electrified line.

- Alternating current (AC) overhead line systems.
- Direct current (DC) third rail systems.
- DC fourth rail systems.

2.6.1 AC overhead line system

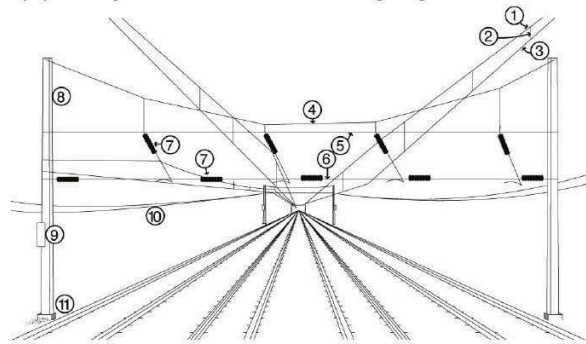


You can only go on or near the line with overhead line equipment if your sentinel card is endorsed with PTS AC.

An example of the typical components incorporated in an overhead line equipment system is shown in the following diagram.

Key

1	Contact wire	7	Insulators
2	Dropper	8	Headspan type structure
3	Contact wire	9	Structure number
4	Headspan wire	10	Return conductors
5	Cross span wires	11	Structure bond
6			



In addition to the wires shown, bare-feeder wires supply electricity from trackside cabins. The overhead line equipment and the attachments are electrified at up to 25,000 volts AC.

Current is taken from the suspended contact wire by a pantograph on top of the train and passed through the train equipment to power the motors.

2.6.1.1 Precautions to take when working on or near to overhead line equipment

- Assume that the overhead line equipment is live at all times.
- You must not use a crane, plant or similar equipment on or near the overhead line equipment unless the:
 - overhead line equipment has been isolated and earthed
 - COSS or SWL has been issued within an overhead line permit.
- Keep people, clothing, tools and equipment at least 2.75 m from:
 - anything attached to, or hanging from, the overhead line equipment
 - any broken or displaced wire connected to the overhead line equipment, whether hanging or lying on the ground.
- Take extra care not to come within 2.75 m of overhead line equipment in cuttings, when working on embankments, structures or in vehicles.
- Keep paint, water and other liquids well away from where they could be thrown, splashed or fall onto live overhead line equipment.
- Keep tools and equipment or ropes, wires, tapes and surveying equipment clear of live overhead line equipment.
- Keep pipes, rods, poles, brooms, mops or ladders horizontal when carried.
- Take special care when standing on the floor or load of open wagons if the overhead line equipment is live on the next line.
- Make sure there is no possibility of branches or debris coming within 2.75 m of live overhead line equipment if trimming or felling trees.
- Use ladders made of wood or approved non-conducting material. Do not use steel or aluminium ladders, or wooden ladders reinforced by metal attachments.
- When cutting water, gas or other metal pipe or metallic cable sheathing you must:
 - first connect a temporary electrical continuity jumper cable across the point where you will make the cut
 - keep the jumper cable in position until the pipe is again complete.



You must immediately contact the electrical control officer (ECO) if a person is in contact, or is in danger of coming into contact, with the overhead line equipment.

2.6.2 DC third rail system

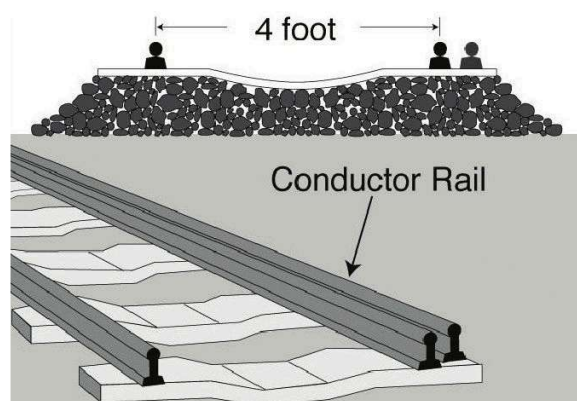


You can only go on or near the line with conductor rails if your sentinel card is endorsed with PTS DCCR.

Current passes through the third rail, which is raised **slightly above the height** of the running rails and is supported on insulators.

The third rail is referred to as the conductor rail, which is carrying 750 volts DC.

Current is picked up from the conductor rail by a pick-up shoe, situated on the outside of the train and passed through the train equipment down to the motors.

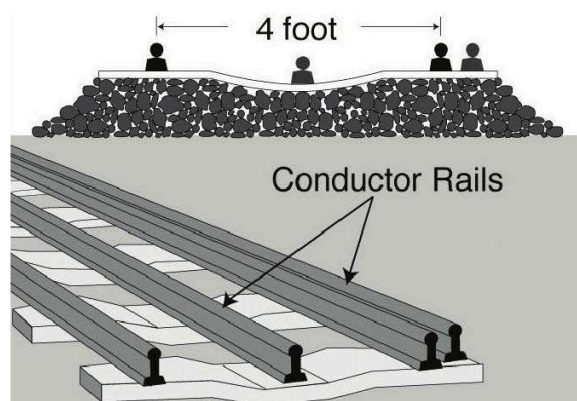


Third rail system

2.6.3 DC fourth rail system

On this system, current is returned to the supply by means of a fourth rail that is situated between the two running rails.

The fourth rail, like the third, is mounted upon insulators, and is slightly higher than the running rail. It is carrying 750 volts DC.



Fourth rail system

2.6.4 Precautions to take when working on or near to conductor rails

2.6.4.1 Precautions before working on or near DC electrified lines

Do not start work on, or close to, the conductor rail (or its connections) until it has been isolated.

The person requiring the isolation will be given a conductor rail permit. However, the issue of this permit does not mean that train movements are stopped on the lines concerned.

Do not work outside the limits shown in the conductor rail permit or those specified by your COSS or SWL.

The only exception is when special authority is given for certain types of work to be carried out without isolation.

In such instances, precautions are specified that may include the use of approved insulated tools, insulated troughing, approved rubber mats or gloves. These must be in good condition and dry.

2.6.4.2 Precautions when working or walking adjacent to live conductor rails

- Keep paint, water and other liquids away from where they could be thrown, splashed or fall onto live conductor rails.
- Keep all tools and equipment or ropes, wires and surveying equipment clear of live conductor rails.
- Keep pipes, rods, poles, brooms, mops or ladders horizontal when carried, to avoid accidentally touching the live conductor rails.

2.6.4.3 Precautions when working or walking near DC lines

- Assume that the conductor rails are live at all times.
- Avoid crossing the tracks unless it is absolutely necessary, and whenever possible cross at a gap in the conductor rail.
- Do not step on, touch or allow clothing, tools or equipment to come into contact with a conductor rail or any of its connections.
- Do not step on protection boarding or between a conductor rail and the adjacent running rail. Instead, step over in one movement.
- Do not touch the collector shoes or their connections on any traction unit, whether or not in contact with the conductor rail.
- Do not step into flood water, which may be in contact with the conductor rail.
- If using a track circuit operating clip or device, apply it first to the rail furthest from the conductor rail.



You must immediately contact the electrical control officer (ECO) if a person is in contact, or is in danger of coming into contact, with the conductor rail.

2.7 Personnel



As far as moving trains are concerned, the best form of personal safety is for every person in the workforce to keep a lookout for themselves and for everyone else.

2.7.1 Communication with the workforce

When working on or near the line the COSS or SWL will brief you on the following.

- The nature and location of the work.
- The approved access point and route to site.
- The limits of the site and how they are defined.
- The lines at site, including the maximum speed and direction of trains.
- If any lines are blocked.
- The best means of contacting the signaller and ECO if necessary.
- Information on any site hazards.
- Whether you will be working in an area where the line is blocked to trains or if the line remains open to trains.
- Any safe limits of the working area, the position of safety and method of warning, if applicable.

On electrified lines this will include the following additional requirements.

- The limits that it is safe to work within.
- Whether any nearby electrification equipment is still live and dangerous and its location.
- Whether any non-electric trains or on track plant (OTP) could still approach on the isolated lines.

When undertaking construction work a task briefing sheet must be used, which will provide the following information.

- | | |
|--|---------------------------|
| ● An outline of the work to be undertaken. | ● Site risks. |
| ● Access and egress arrangements. | ● Emergency arrangements. |
| ● Activity risks. | ● Contact information. |
| ● Resources. | |

At the end of the briefing operatives should be asked to confirm their understanding. At this point the COSS or SWL should ask some questions to confirm understanding.

Operatives should be given the opportunity to ask any questions too.



Operatives should be told not to sign the confirmation of the briefing unless they are clear on the safe system of work.

2.7.2 Personal protective equipment (PPE)

The main priority is that workers are easily seen, particularly by train drivers. Therefore, wearing high-visibility clothing is an important requirement. The clothing must conform to the requirements laid down by Network Rail and must be clean and worn correctly.

However, train signals are in three colours: red, yellow and green, and so to reduce confusion to train drivers, there are colour restrictions for items of clothing worn by people working on or near a railway.

Red, green or yellow clothing must **not** be worn in case a train driver mistakes the item for a signal. Railway workers should wear fluorescent orange.

The minimum PPE for the work site is shown below.

- High-visibility upper body clothing with reflective tape, which complies with EN ISO 20471:2013 Class 2 and RIS-3279-TOM.
- High-visibility lower body clothing with reflective tape, which complies with EN ISO 20471 Class 1 and RIS-3279-TOM.
- A safety helmet, which complies with BS EN 397:2012.
- Safety footwear, which complies with BS EN ISO 20345:2011, provides support to the ankle, includes mid-sole protection and has a protective non-conductive toecap.
- Other items of PPE as may be required to safely carry out specific tasks.



All high-visibility clothing must conform to the requirements laid down by Network Rail

2.7.3 Working hours

At the Clapham Disaster Inquiry, Lord Justice Hidden made several recommendations regarding the causes of accidents. One of his recommendations was that the hours of work should be controlled to prevent excessive times on duty.

As a result of these recommendations, the contractor must now prohibit excessive overtime working and shift lengths. Accordingly, personnel should:

- not work more than 13 turns of duty within any 14 consecutive days
- not work more than 72 hours within seven consecutive days
- have a minimum rest period of 12 hours between consecutive shifts
- not work more than 12 hours in any one shift.

Where a contractor is employed to work on signal and control systems or the renewal or maintenance of track, the requirements are enhanced to not more than 13 turns of duty to be worked in any 14-day period, and no more than 23 turns of duty to be worked in any two consecutive 14-day periods.

A contractor is required to keep records of the working hours of all staff working on railways, and these records must be kept available for inspection and audit.

2.7.4 Drugs and alcohol

The Transport and Works Act 1992 stipulates the limits for alcohol consumption in the rail industry.

This legislation lays down the methods that must be adopted to ensure that anyone working on the railway is not under the influence of drugs or alcohol.

Personnel, both railway staff and contractors, working on a railway will be medically screened prior to employment for the presence of any of the following.

- Drugs, other than medication that does not affect their work performance.
- More than 29 milligrams of alcohol in 100 ml of blood.
- More than 13 micrograms of alcohol in 100 ml of breath.
- More than 39 milligrams of alcohol in 100 ml of urine.

Compare these figures to those laid down for driving a car.

- Not more than 80 milligrams of alcohol per 100 ml of blood.
- Not more than 35 micrograms of alcohol in 100 ml of breath.
- Not more than 107 milligrams of alcohol in 100 ml of urine.

It can be seen that far stricter controls are placed on those working on the railway infrastructure than on the ordinary car driver.

There is a 'for cause' screening system for screening after a safety-critical incident has taken place, where drugs or alcohol may be considered a contributory factor. All personnel working on the railway must comply with screening.

Personnel found to register positive in any screening will be dismissed and could face prosecution under the Transport and Works Act for working under the influence of drugs or alcohol in a safety-critical work post.



All railway workers are subject to unannounced random screening in the workplace. This may take place at any time.

Appendix A – Train sighting chart

Required sighting distances are calculated from the following chart.



The warning time must be sufficient to enable everyone to be in a position of safety at least 10 seconds before the arrival of a train.

Permissible speed	Sighting distances (metres) to provide minimum warning time of:						
(mph)	15 sec	20 sec	25 sec	30 sec	35 sec	40 sec	45 sec
125	900	1,200	1,400	1,700	2,000	2,300	2,600
120	900	1,100	1,400	1,550	1,900	2,200	2,500
115	800	1,100	1,300	1,550	1,800	2,100	2,400
110	800	1,000	1,300	1,500	1,800	2,000	2,300
105	800	1,000	1,200	1,450	1,700	1,900	2,200
100	700	900	1,200	1,350	1,600	1,800	2,050
95	650	850	1,100	1,300	1,500	1,700	1,950
90	650	850	1,050	1,250	1,450	1,700	1,850
85	600	800	950	1,150	1,350	1,600	1,750
80	550	750	900	1,100	1,300	1,500	1,650
75	550	700	850	1,050	1,200	1,400	1,550
70	500	650	800	950	1,100	1,300	1,450
65	450	600	750	900	1,050	1,200	1,350
60	450	550	700	850	950	1,100	1,250
55	400	500	650	750	900	1,000	1,150
50	340	500	600	680	800	900	1,050
45	320	420	520	620	720	820	920
40	280	360	460	540	640	720	820
35	240	320	400	480	560	640	720
30	220	280	340	420	480	540	620
25	180	240	280	340	400	460	520
20	140	180	240	280	320	360	420
15	120	160	180	220	240	280	320
10	80	100	120	140	160	180	220
5	40	60	60	80	80	100	120



The permissible speed at a location is 75 mph, and the work is of such a nature that the minimum warning time required is 30 seconds. Therefore, the sighting distance required is 1,050 m.

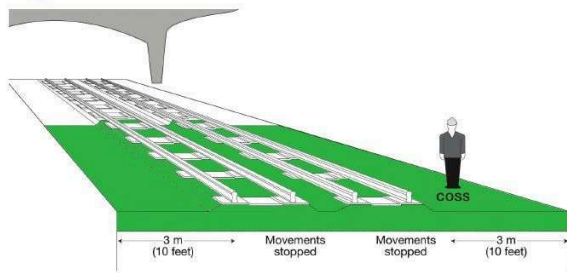
However, not all lengths of track are straight and the view may be obstructed by bends, bridges or other structures. Therefore, it is essential for the COSS to assess these potential hazards as part of the risk assessment and by possibly appointing more than one lookout, or alternative mechanical means operated by competent users, in order to obtain the necessary sighting distance for an approaching train and to allow time for everyone to get into a position of safety at least 10 seconds before the arrival of the train. There will need to be a warning of the train's approach, which will be given by the site lookout.

If the work involves the use of noisy equipment, the noise produced may prevent the work team from being able to hear any warnings given by lookouts, a problem compounded by the requirement of wearing hearing protection.

In such situations, a system of alerting the workers by the use of a touch lookout should be implemented. This entails a lookout at the worksite to touch all the members of the work team to warn them of approaching trains.

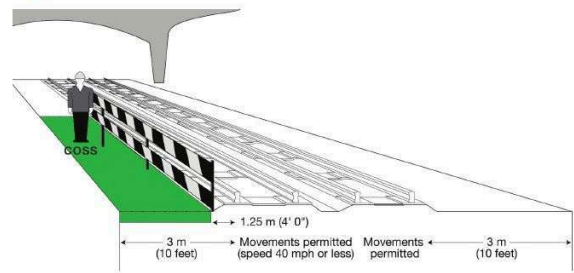
Appendix B - Safe system of work hierarchy

Figure 1



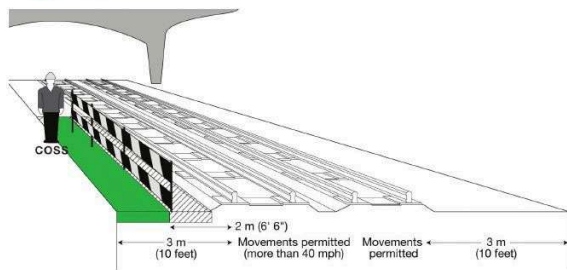
The diagram above shows a safeguarded safe system of work. This means train movements are stopped on all lines. Be aware that track plant may be used on any line.

Figure 2



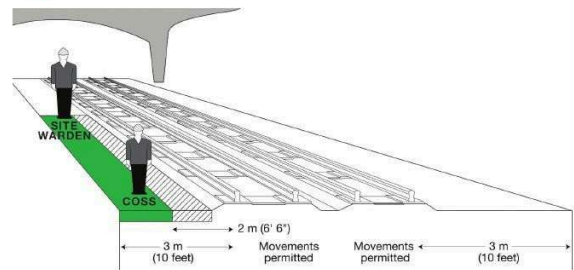
The diagram above shows a fenced safe system of work. This example is of a rigid safety barrier, which should be positioned no less than 1.25 m from the nearest open line.

Figure 3



The diagram above shows a fenced safe system of work. This example is of a plastic netlon safety barrier, which should be positioned no less than 1.25 m from the nearest open line if trains are travelling no more than 40 mph. If trains are travelling more than 40 mph the distance must be increased to 2 m.

Figure 4



This diagram shows a safe system of work using site warden warning. The site warden warning area must be at least 2 m from the nearest open line. If you stray outside of the site warden warning area, the site warden will shout a warning: 'Get back'.

If none of the above safe systems of work can be employed then an equipment warning safe system of work may be used, which will warn you when a train is coming.

The final and last resort safe system of work is lookout warning. This safe system of work uses unassisted lookouts. The warning is given by a horn, whistle and touch (if there is noise or the group are wearing hearing protection).

In both of these instances, the COSS or SWL will make sure you get enough warning to enable you to reach a position of safety at least 10 seconds prior to the arrival of a train.

Appendix C – Good practice guide on competence development

Purpose

The *Good practice guide (GPG) on competence development* represents a collation of previous Rail Safety and Standards Board (RSSB) good practice guides relating to training and competence, and reflects the latest developments and thinking around how to ensure staff competence.

It is designed to be used as a reference guide and a source of inspiration to help people working in the area of competence, allowing them to pick and choose ideas that are useful and practical, depending on their own requirements.

The GPG aims to move its users past the minimum standard as required by legislation – it is about going beyond the basics and providing the tools needed to develop comprehensive competence management systems as per your business and individual needs.

As understanding in the field of competence is always evolving, this is a live document that will be periodically reviewed and updated to reflect the latest thinking and resources relating to competence development.

Target audience

Railway duty holders (infrastructure managers and railway undertakings) are required under the ROGS Regulations to make provisions within their safety management system (SMS) to ensure, so far as is reasonably practicable, that the competence of all safety-critical staff under their control is developed and maintained to a minimum safe standard.

This GPG has been developed to provide information and practical guidance to anyone who needs to understand, manage or contribute constructively to the analysis, design, delivery, review or assessment of training and other competence management activities.

The core text of this GPG is relevant to anyone with an interest in competence development (including the learners themselves), whereas the appendices are for more specific audiences (such as those with a more detailed interest in simulators, assessments, etc.).

Scope

This guide is designed to act as a reference manual. It pulls together:

- the **evidence-based findings** of recent research projects, highlighting **practical guidance** rather than focusing on the academic theory behind them
- the content of the **previous five RSSB good practice guides** in the area RS/220; *Good practice in training* (RS/220)*, *Good practice guide to train driver training* (RS/221)*, *Good practice on simulation as a tool for training and assessment* (RS/501)*, *Good practice guide on competence review and assessment* (RS/701) and *Good practice guide for driver assessment* (RS/702)
- **details of technology** or other relevant mediums that support the implementation of innovative concepts
- a variety of small and more in-depth **case studies** (these provide real-life demonstrations to highlight good practice in the various areas, illustrating how different companies have adopted approaches that suit their learners and business)
- **links and signposts** to practical tools and further information.

The advice in this GPG mainly pertains to the competence of safety-critical staff, though many of the principles are widely applicable in the field of learning and development and will have relevance to other roles. Although the overriding objective of competence management is mitigation of safety risk, it should be borne in mind that safety risk is one component of overall business risk.

**This good practice guide is not a mandated document, even though it sits in the suite of Railway Group Standards documents.*



For further details refer to the Rail Safety and Standards Board's *Good practice guide on competence development*.

(The above text is extracted from the Good practice guide on competence development, Document no. RS/100, Issue 1, March 2013. This document is copyrighted and has been reproduced with the permission of the Rail Safety and Standards Board Ltd.)

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GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

Overview

Demolition can be a hazardous activity. It is technical, can be high in complexity and should only be carried out by appropriately trained, competent and authorised personnel.

This chapter gives an understanding of the potential problems associated with demolition, the good practices that must be followed and the requirement and importance of appointing competent demolition contractors.

3.1 Introduction

Safe demolition is a process that requires technical skill and competence. It can be hazardous if carried out by contractors who are not experienced and competent in demolition planning, management and delivery.

The word *demolition* is often misinterpreted to only mean large-scale, or to involve the complete removal of a structure or building. However, the removal of a single element or the partial removal of a structure or building also constitutes demolition.

Information, planning, implementation of suitable and sufficient controls, and the provision of adequate and appropriate resources are essential elements for safe and successful demolition projects.

Whether demolition is required for a small building or structure, or for a complex industrial, commercial or residential site, you must be aware of the hazards and risks. These need to be identified, assessed and effectively controlled to avoid danger and remove or reduce the potential for injury to persons and damage to adjacent property and the environment, as far as is reasonably practicable.

A systematic approach to the demolition process starts with responsible clients making suitable arrangements for managing a project, including making sure duty holders are appointed and sufficient time and resources are allocated. Inexperienced clients may need to seek the assistance of a demolition engineer or consultant for guidance in the appointment of duty holders and management of the demolition process.

- Demolition works should be planned in such a way that they are undertaken using systems of work that take into consideration safety, health, the environment, cost and efficiency.
- All those involved, from concept, through tender, to contract delivery, should ensure effective planning and communication, through the provision of sufficient time and resources to facilitate effective planning and delivery of a safe, successful and efficient project.
- Time is required to acquire licences, permits and to disconnect utility services. Utility surveys should start in the pre-construction phase.
- After the contract has been awarded, and before work commences, the demolition contractor should determine the proposed safe sequence of operations on the basis of an assessment of the comparative risks, related specifically to the site and local conditions.
- Pre-construction information, the sequence of operations, programme and safe system of work should form the basis of the construction phase plan, which will include the arrangements for carrying out demolition or dismantling. These arrangements must be written down before the work starts.
- The methods of work should allow for demolition work and site clearance, as stipulated by the contract, taking account of any constraints imposed by the client or local conditions.
- Adequate time should be allowed for setting up the site, including the provision of welfare facilities, prior to work commencing.
- Waste management plans should also be developed.

3.2 Important points

- Demolition can be hazardous and should only be entrusted to experienced contractors who employ well-trained, competent personnel.
- Many demolition projects, particularly within buildings constructed prior to 1999 (especially between 1960 and 1990) may contain asbestos.
- Recognise that structural refurbishment may involve demolition work, which should be planned with the appointment of a competent contractor.
- Planning and supervision is vital to ensure a successful project.
- Demolition should be carried out with due regard to the British Standard (BS 6187:2011), good practice current at the time the works are being carried out, Health and Safety Executive (HSE) and industry guidance, and in full compliance with regulations.
- Ensure that prior to all demolition a site-specific demolition asbestos survey, as defined in the HSE publication *Asbestos: the survey guide* (HSG264), has been carried out by a competent person and that the information is available to those who need it.
- It should be noted that the HSE advises wherever paintwork or coatings are likely to be damaged or disturbed a lead (paint) survey should be carried out, regardless of the age of the building or the extent of the works.

- **The Construction (Design and Management) Regulations 2015 (CDM) apply to all demolition and dismantling projects,** regardless of size, but not all projects will be notifiable to the HSE.
- Ensure the works are properly planned, follow a safe system of work with written risk assessments and method statements developed, communicated to the personnel carrying out and supervising the work and understood by all concerned.
- Managers should ascertain the stability of the structure due for demolition. Where there are any doubts or concerns, the services of a demolition engineer and/or structural engineer should be engaged.
- The removal of non-structural fixtures, fittings and finishes, often referred to as soft stripping, must be planned and controlled to promote reuse or recycling and to avoid the risk of personal injury, uncontrolled release of asbestos, falling objects and so on.
- Demolition arisings should follow the hierarchy of waste minimisation (for example, reuse, recycle and recover) with waste being handled, managed and disposed of in accordance with current legislation and the waste duty of care Code of Practice.
- Demolition and refurbishment information datasheets (DRIDS) provide informative and practical guidance around reuse or recycling of the materials and products found on modern demolition sites.
- If processing and disposing of materials arising from the demolition (such as screening or crushing), ensure the relevant licences, permits or exemptions are in place.
- Ensure the necessary waste documentation is obtained, completed correctly and records kept.
- A waste management plan should be developed and kept up-to-date, including records of all demolition waste streams and the quantities. The plan will identify opportunities for materials to be reused, reclaimed and recycled, minimising the amount of waste being sent to landfill.



Code of Practice for full and partial demolition (BS 6187:2011)

The code of practice gives good practice recommendations for the demolition (both full and partial) of facilities, including buildings and structures. This standard applies to demolition activities undertaken as part of structural refurbishment. It also covers decommissioning.

The standard gives recommendations for the following.

- The proper and effective management of demolition processes, including those forming part of structural refurbishment.
- Maintaining structural stability, including through the provision of temporary structural support, where necessary.
- Managing deliberate structural collapse.
- Identifying and establishing responsibilities during all phases of the demolition process.
- Acquiring a knowledge of the site, including its former uses.
- Managing environmental issues.
- Managing health and safety hazards.
- Carrying out risk assessments, developing safe systems of work and planning the work accordingly.
- Establishing and managing procedures effectively.
- Determining and managing safe exclusion zones.

It is essential that those carrying out demolition activities possess the necessary levels of competence. An important aspect of the code of practice is reference to exclusion zones and the safety of both people on site and others who may be affected by the work.



For further information refer to guidance from the National Federation of Demolition Contractors (NFDC).



For general definitions and terms used by the demolition industry refer to Appendix A.

3.3 Legislative requirements

3.3.1 The Building Act 1984

Section 80 of the Act requires persons intending to carry out the demolition of a structure, or part-structure greater than 1,750 cubic feet (approximately 50 m³), to give six weeks' notice to the Local Authority Building Control department. This may incur an administration fee. A Section 81/82 counter notice is usually issued, providing conditions that apply to the works or a refusal with reasons. This is normally something that the client or design team should have carried out, but it frequently falls to the demolition contractor.

3.3.2 Construction (Design and Management) Regulations 2015 (CDM)

Readers are advised:



- to refer to the industry guidance that supports CDM
- that under these regulations the definition of construction includes demolition and dismantling, and is to be regarded as such when used in the following text.

CDM applies to all construction, which includes demolition and dismantling work, as defined in BS 6187, regardless of the project's size or duration. A construction phase plan must be developed for all construction projects. A principal designer (PD) and principal contractor (PC) must be appointed where more than one contractor is engaged. Where there is only one contractor engaged a contractor and designer must be appointed by the client. This contractor (demolition contractor) will be responsible for health and safety on the site and for the development of the construction phase plan.

A project is notifiable if the construction work on a construction site is scheduled to last longer than 30 working days and have more than 20 workers working simultaneously at any point in the contract, or exceed 500 person days.

To ensure site safety during demolition the principal contractor or contractor must ensure that one or more competent person(s) are appointed to plan and supervise the work. The level of supervision required will depend upon the risks to health and safety and the skills, knowledge, training and experience of the workers (for example, workers will require closer supervision if they are young, inexperienced or starting a new work activity). Other factors to consider are the levels of the individuals' safety awareness, education, physical ability, literacy and attitude. Any duty holder who puts another person to work on a construction site (for example, a principal contractor engaging the services of a demolition contractor) must ensure that the person is competent to do what is required of them. Any person engaged must ensure that they have the competencies, skills and experience required for the work, otherwise they must refuse the work.

The NFDC and the Institute of Demolition Engineers (IDE) can provide information regarding the competence required for particular projects.



A construction phase plan is required for all projects.

3.3.2.1 Stability of structures

- All practicable steps shall be taken to prevent danger to any persons, to ensure that any new or existing structure or any part of such structure that may become unstable or in a temporary state of weakness or instability, due to carrying out of construction work, does not collapse.
- Any buttress, temporary support or temporary structure must be designed, installed and maintained so as to withstand any foreseeable loads that may be imposed on it, and must only be used for the purposes for which it is so designed, installed and maintained.
- No part of a structure shall be so loaded as to render it unsafe.



For further information refer to Chapter C04 Temporary works.



For further information refer to the *Code of Practice for temporary works procedures and the permissible stress design of falsework (BS 5975)*.

3.3.2.2 Demolition or dismantling

The demolition or dismantling of a structure or part of a structure shall be planned and executed to reduce risk to as low a level as is reasonably practicable. The arrangements for carrying out demolition or dismantling work must be recorded in writing, prior to the commencement of the work.

3.3.2.3 CDM duty holders

The **client** is responsible for the provision of pre-construction information, which is already in their possession or is reasonably obtainable by them (such as the results of any refurbishment or demolition asbestos survey and the location and status of utility services). The client must also ensure the following.

- The project is notified to the HSE, if it meets the criteria defined in BS 6187.
- That adequate welfare facilities are provided.
- Relevant parts of the pre-construction information are distributed to contractors and designers to enable them to tender for, or prepare to carry out, the work.
- Suitable arrangements are made for managing the project so that health, safety and welfare are secured.



The client sets the tone for the intended demolition and, as such, care should be taken to ensure sufficient time and resources are made available to properly plan, procure and carry out the works.

The **principal designer** must plan, manage, monitor and co-ordinate the work of the pre-construction phase. They must liaise with the principal contractor in providing relevant information for the planning and monitoring of the construction phase.

Principal designers may also be designers. Design work may include any structural work and temporary works design.

The principal designer is responsible for the co-ordination of health and safety design and planning aspects, including those below.

- Co-ordinate matters relating to health and safety during the pre-construction phase.
- Estimate the period of time required to complete work stages.
- Take account of general principles of prevention.
- Identify, eliminate or control foreseeable risks to health and safety of any person carrying out construction work, maintaining or cleaning a structure and/or using a structure designed as a workplace.
- Ensure designers comply with their duties.
- Ensure all persons co-operate with the client, principal designer and each other.
- Assist the client in the provision of the pre-construction information.
- Provide pre-construction information promptly and in a convenient form to every designer or contractor.
- Liaise with the principal contractor and share information relevant to the planning, management and monitoring of the construction phase and the co-ordination of health and safety during construction.

The minimum lead times for the project must form part of the pre-construction information. Principal designers must themselves be competent; demolition and asbestos issues may be too specialist for many who normally perform the principal designer's role and they should be prepared to take specialist advice, if required.

Depending on the number of contractors on site, the **principal contractor** or the **contractor** is responsible for the overall construction phase, taking into account health and safety issues and the development of the construction phase plan.

Contractors and other workers must all work together as a team to achieve high standards in health, safety and welfare and protection of the site. The contractors, which normally include utility companies, must co-operate with the principal contractor by obeying site rules and complying with relevant parts of the construction phase plan.

Regardless of the size of the contract, the client has to ensure there are suitable management arrangements for the project welfare facilities and allow sufficient time and resource for all stages. A written **construction phase plan** will be required for all demolition projects. The plan should be written by the contractor and include training and information for workers.

The **demolition contractor** may be appointed as the principal contractor subject to having the necessary expertise, understanding and resources to undertake the role.

Irrespective of the size of the project, the demolition contractor must ensure the work is properly planned and carried out safely with the arrangements recorded (risk assessment, method statements, programme, construction phase plan and so on). They must also ensure these arrangements are communicated and clearly understood by those carrying out the work.

3.3.3 Provision and Use of Work Equipment Regulations 1998

The Provision and Use of Work Equipment Regulations (PUWER) apply to work equipment used in demolition. They require that an employer supplies work equipment that is safe, correct and suitable for the job, the equipment is maintained and staff receive information, instruction and training in the use of the equipment.

Demolition can involve some work being carried out at height, particularly salvage and soft stripping (for example, the recovery of slates), which has in the past been the cause of deaths and injuries to demolition operatives. Due to advances in demolition techniques, and the increasing size and reach of the machines used, there is now a reduced need to work at height.

However, the presence of such machines on site brings about responsibilities for maintenance, inspection, thorough examination, operator competence and so on under these and other regulations.

Under the regulations, falling object protective structures (FOPS) are required for machinery. A demolition specification machine must have a FOPS-rated cab, as it is foreseeable when working overhead that material could fall onto the cab. Demolition attachments are common on a demolition site.

Quick hitch couplers requiring the fitting of a safety pin when changing attachments should never be operated without that pin being in place. The use of reinforcing bars as makeshift retaining pins for attachments is not acceptable and would be contrary to these regulations.

It is common to see mobile crushers employed during a demolition project to process the demolition arisings for reuse as part of ongoing works on site. These machines can pose a great risk to untrained or unaware operatives. Issues such as machinery guarding and clearing blockages, as well as dust, noise and vibration, must be considered.

Safe working practices dictate that operators should avoid standing on a crusher whilst it is running (for example, trying to remove foreign objects from the demolition arisings) or near the crushing jaws.

www For further information refer to the NFDC guidance on demolition attachments.

3.4 Project documents

3.4.1 Pre-construction information

Relevant pre-construction information, relating to the health, safety and environmental hazards associated with the work, should be provided by the client regardless of the size of the project. Where more than one contractor is engaged, the principal designer must carry out the tasks outlined previously, with regard to the collection and distribution of the pre-construction information.

In the context of this chapter, such information is likely to include:

- the location and physical condition of any asbestos where its presence has been identified by a refurbishment or demolition asbestos survey, as defined in HSG264
- the results of structural surveys
- notices, acknowledgements and disconnection certificates received for all services
- plans identifying the location of underground and overhead services
- the possible presence of contaminants, for example:
 - polychlorinated biphenyls (PCBs) in old electrical transformers
 - the residual contents of tanks and pipelines
 - the location and nature of contaminated ground
- environmental information on watercourses, wildlife, archaeology, and so on
- any other information that is required to ensure that the work can be planned safely and would not reasonably be expected to be known by a competent contractor.

After being appointed by the client, the principal contractor must use the pre-construction information provided by the client to develop the construction phase plan to the satisfaction of the client.

3.4.2 Construction phase plan

The construction phase plan provides a focus for the construction phase of a project.



The arrangements for carrying out demolition, or partial demolition, should be recorded in writing (with a level of detail proportionate to the risks involved) before the demolition work begins. The demolition plan should form part of the construction phase plan.

The construction phase plan must be:

- sufficiently developed prior to any work commencing on site
- reviewed and amended as often as is necessary for the duration of the project
- communicated to those on the site insofar as it is necessary for any person to be aware of any part of it.

3.4.3 Health and safety file

The health and safety file is a record of information for the client or user of the premises. Its purpose is to inform anyone who might be responsible for the structure or site, now or in the future, of the risks that will have to be managed during any maintenance, repair or renovation. Generally, for demolition work, the health and safety file will contain information such as details of the following.

- Any services that have been capped or discovered and worked around.
- The presence of any voids and details of any areas that may have been filled.
- The actions taken to remove or treat contaminated land.
- Asbestos removal from retained structures, including copies of the clearance certificate and disposal records.

Such information may be useful to add to the asbestos management plan for a site, or to provide confirmation for future use that contaminants have been removed from the site. A health and safety file is only required for projects involving more than one contractor, although the client may insist on one being developed for future reference. Initially the health and safety file is the responsibility of the principal designer but if the duties of the principal designer finish before the end of the project then responsibility passes to the principal contractor.

3.4.4 Programme

The programme time allocated must be adequate to allow the demolition work to be planned and carried out safely. This should detail the mobilisation and planning, and correct sequence of the works with any hold and review points. Where the programme has unavoidable interfaces with other trades or contractors on site, this must be recognised as a potential area of risk. Good communications, planning and management are therefore essential to ensure high standards in health and safety. The operator of mobile crushing equipment must also give notice to the relevant Local Authority of their intention to carry out crushing work each time the equipment is moved.

3.5 Essential elements of a construction phase plan

When developing a construction phase plan that incorporates demolition work, the following elements must be included (also refer to *Chapter A03 Construction (Design and Management) Regulations 2015, 3.8.2 Construction phase plan*).

3.5.1 Project information

Names, addresses, contacts and telephone numbers should be detailed indicating the client, project managers, quantity surveyors, architects, structural engineers, the principal designer and principal contractor.

3.5.2 Scope of work

An explicit and concise opening paragraph should be included, outlining the extent of work, along with any other related work (such as asbestos removal), items or equipment to be recovered and set aside, façade retention, and so on.

3.5.3 Existing environmental information and drawings

Where available, construction drawings should have been supplied to the contractor, by the client, along with information on the former use of the site or buildings to be demolished. This information should include the health and safety file for the premises (if available), to determine what (if any) physical or chemical hazards are on the site or in the buildings (such as underground tanks and/or potential chemical or biological hazards or contaminated land).

The details of the building or structure should be checked to ensure that it is not a listed building. In addition, the locality of the site in relation to schools, roads, hospitals, protected wildlife, plants, watercourses and so on, which may be affected by the demolition activity, should be taken into account.

3.5.4 Risk assessment and special hazards

Having gathered all available information and visited the site to assess the work involved, along with identifying all known hazards and confined spaces, the contractor's appointed person on site should ensure that risk assessments (such as assessments for asbestos, lead, hazardous substances or noise) are made. Method statements should then be drawn up.

Information from these assessments should be included in the construction phase plan for the project, which must also take into consideration risks to health, and detail any safety precautions that are being taken to control and manage the risks.

Consideration must be given at an early stage to control the access for vehicles and pedestrians (including any visitors) and, where practicable, ensure separate access and egress for vehicles and pedestrians.

Finally, assess the impact that the site environment will have on any people who might be affected by the activity (such as neighbours or members of the public).



For further information on method statements and risk assessments, including examples of both, refer to Chapter A05 Risk assessments, method statements and permits to work.

3.6 Selection and appointment of a demolition contractor

The client or principal contractor, in selecting a demolition contractor, should satisfy themselves of the contractor's competence, knowledge, ability and resources to carry out the work safely.

Expertise is vital. If there is more than one contractor on site, the client has to appoint a competent principal designer and principal contractor to co-ordinate and manage the health and safety issues during the demolition works.

The principal contractor's role can be taken by the demolition contractor. References must be sought, and relevant questions asked, to establish the credibility of the contractor involved. If there is only one contractor on site, there is no requirement to appoint a principal contractor or principal designer.

The contractor and designer(s) respectively will have to fulfil their duties in accordance with CDM.

Care must be taken to ensure that the principal designer has the necessary level of competence in demolition work as it is a highly specialised field.

Alternatively, it may be wise to consider appointing an independent demolition engineer as an adviser.



A useful source of information is the National Federation of Demolition Contractors (NFDC).



When looking to appoint a competent demolition engineer, membership or fellowship of the Institute of Demolition Engineers (MIDE/FIDE) and/or the Construction Skills Certification Scheme (CSCS) professionally qualified person (PQP) in demolition is a good indicator of their competence.

3.6.1 Pre-qualification

If more than five people are employed by the demolition contractor, then that contractor should have a written health and safety policy.



For further information refer to Chapter A04 Health and safety management systems.

The health and safety policy should be signed and dated by the director responsible and communicated to employees.

The client should also consider the following.

- Does the contractor employ a health and safety adviser, or do they purchase the services of a health and safety adviser or consultants?
- Does the contractor employ a demolition engineer, or do they purchase the services of demolition consultants?
- If consultants are used, how often will they visit the site?
- Are the consultants experienced in the health and safety management requirements of demolition sites?
- Are the consultants always available for advice, particularly if something goes wrong?

The safety record of the contractor should also be considered.

- Are safety statistics for injuries, lost time and dangerous occurrences kept and available?
- Has the HSE issued any improvement or prohibition notices against the contractor? If so, have the root causes of such been resolved?
- Are there any pending potential enforcement actions against the contractor?
- Is it the intention to use sub-contractors? If so, for what elements and how do they select and control sub-contractors?
- Is the contractor a member of the NFDC?

Membership of the NFDC is a useful competence indicator and will give the client the assurance of a minimum of five years' satisfactory trading experience, and a knowledge that certain recognised minimum standards have been and are being met (for example, insurance, training and supervision).

However, irrespective of membership of the NFDC, checks must be made regarding whether the organisation has the relevant experience to be deemed competent for the project in accordance with the requirements of CDM.

- Does the contractor have adequate public and employer's liability insurance that covers demolition?

Check for any restrictive covenants or other restrictions (for example, for work to be carried out above a height of 30 m).

- Do the company's employees hold the appropriate certification of competence for demolition operatives (CCDO) cards for operatives, supervisors and managers (Levels 1, 2 or 3)?
- Are references available? It is useful to speak to a client from a similar project size and complexity, as well as from their last project.

3.6.2 Appointing a contractor

A genuine and competent demolition contractor will not object to supplying the name of the client for whom the last contract was carried out. If that client is then contacted, they can be asked whether they were satisfied with the work.

They can be asked for information on the works, type, size, approximate value and so on, and asked the following questions.

- Were any problems experienced during the works?
- Would they use the contractor again?
- How did the contractor perform overall?

These are some examples of the kinds of question that should be asked to ensure satisfaction when considering the appointment of such a contractor. An interview with the potential contractor should be held to clarify matters. The client can also see if they would be prepared, or if they are qualified and competent, to take on the role of principal contractor.

3.7 Planning for demolition



It cannot be emphasised too strongly that demolition is potentially hazardous and must be left to the experts (competent persons). Experience, knowledge, training, planning and supervision are of paramount importance.

Before any work starts, the implications of the demolition project to be carried out must be determined.

- What is the age of the building(s) and what was (were) the previous use(s)?
- What is the scope and extent of the works?
- What is the type of construction and what were the materials used?
- Is it full or partial demolition?
- Where are the separation points, if any, and what are the perceived load paths?
- Are floor slabs, foundations or piles to be removed?

- Are there any hazardous substances or invasive species in, around or under the building that is to be demolished (for example, asbestos, lead paints, flammable liquids, unidentified drums or packages, Japanese knotweed and so on)?
- Is asbestos suspected? (If so, a full invasive demolition survey must be carried out to form part of the pre-construction information.)
- Is the building on contaminated land?
- Are there any proximity hazards present (such as overhead or underground services, voids, tunnels, watercourses and so on)?
- Are there any site restrictions (such as access and egress)?
- Are there people and adjacent properties that may be affected by the proposed working hours?
- Will people be affected by noise or vibration emanating from the site? If so, a Section 61 noise notice may be required.
- Will there be a need to carry out a dilapidation survey of the adjacent buildings and/or highways?
- Will any waste management licences or permits be required to process or reuse the demolished material?
- Is the project likely to be notifiable?
- Who will notify the Local Authority under the Building Act, Section 80?

The information above would form the basis of the pre-construction information collected by the client, and assisted by the principal designer, under CDM. Once these things have been determined, decisions should be taken as to what are the acceptable or unacceptable methods to carry out the demolition.

Information on contaminated land must include details of the past and current uses of the site. Advice can be sought from the Environment Agency (EA), Northern Ireland Environment Agency (NIEA), Natural Resources Wales (NRW) or the Scottish Environment Protection Agency (SEPA), and the Local Authority Environmental Health Department. This may result in a requirement for a site survey, including the taking, observation and analysis of core samples.

Two important final questions remain.

- What is the timescale for the project and is it sufficient for the demolition to be carried out safely?
- How should the job and site be left safe?

The first point is particularly relevant as CDM place a duty on the client to indicate the minimum project mobilisation period.

3.7.1 Personnel and the appointed person

All persons on site are responsible for health, safety and the environment. The following have specific responsibility for health and safety and for supervision of the site.

- All designers.
- The principal contractor.
- The demolition contractor and specialist sub-contractors (for example, asbestos removal contractors).
- The director responsible for safety.
- The site safety adviser.
- The contracts manager.
- The environmental manager.
- The appointed person – a competent demolition supervisor.
- The person who has first-aid training or emergency first-aid training.
- All other contractors on site.

A method of recording who is on site must be rigorously maintained throughout the contract by signing in and out in the (daily) site register. The principal contractor or contractor, with support from the appointed person and others, must ensure that the relevant sections of the construction phase plan are communicated to all people on site at appropriate intervals (such as site induction, task briefings and toolbox talks). These occasions should be recorded and signed for in order to indicate ownership by the recipients.

White card	NDTG approved trainer: five year renewable.
Yellow card	Demolition site visitor: five year renewable.
Green card	Demolition labourer L1 equivalent: five year renewable.
Red cards	Demolition apprentice: two year non-renewable. Demolition and refurbishment operative: three year non-renewable, NVQ required. Demolition topman: three year non-renewable, NVQ required. Demolition chargehand: three year non-renewable, NVQ required. Demolition supervisor: three year non-renewable, NVQ required. Demolition manager: three year non-renewable, NVQ required.
Blue cards	Demolition and refurbishment operative NVQ L2: five year renewable. Demolition topman NVQ L2: five year renewable.
Grey card	Demolition chargehand NVQ L3: five year renewable.
Gold card	Demolition supervisor NVQ L3+: five year renewable.
Black card	Demolition manager NVQ L6: five year renewable.



Company fined £100,000 and director given suspended prison sentence after worker dies

A property development company has been fined £100,000 and a building firm director handed a suspended prison sentence after a labourer was fatally crushed while demolishing a wall.

The self-employed labourer was hired as a sub-contractor by the building firm to work on a house refurbishment project for the property development company in Liverpool. The 41-year-old, who was originally from Czechia, was tasked with demolishing a rear yard wall dividing the property from the neighbouring house.

That same day, other workers on the job left the site at 3.30pm; however, when a neighbour returned home from work at around 5.40pm they saw the labourer trapped between an outer kitchen wall and a collapsed section of the yard wall. He was pronounced dead at the scene by emergency services.

An investigation by HSE found that the demolition work was not planned nor accounted for in the construction phase plan. No risk assessment or method statement was provided, and the labourer was not trained to carry out safe demolitions.

There was also a lack of supervision, as the system of work implemented by the property developer and the building company prohibited non-English-speaking workers from carrying out demolitions.

The development company from Liverpool pleaded guilty to breaching Regulation 4(1) of CDM. They were fined £100,000 and ordered to pay £8,401.59 in costs at Wirral Magistrates' Court.

A director of the building company, from Holyhead, Anglesey, pleaded guilty to breaching Regulation 13(1) of CDM, by virtue of 37(1) of the Act, and was given 26 weeks' imprisonment, suspended for two years, and ordered to pay £5,836 in costs at Wirral Magistrates' Court.

A HSE inspector said: 'HSE will not hesitate to consider the roles of not only organisations when investigating serious incidents, but also those individuals such as directors and managers who are the controlling minds and best placed to direct work and ensure that it is carried out without risks to health and safety.'

(Source: HSE.)



A card scheme for the certification of competence of demolition operatives (CCDO) is managed by the National Demolition Training Group.

3.7.2 Services

Before any work starts, all utility companies must be contacted by the client or, if agreed, by the principal contractor, and sent a site plan showing the footprint and extent of the planned demolition, with the proposed commencement date, requesting the disconnection or isolation of the appropriate service (for example, electricity, gas, water, telecommunications or other cables).

These requests should be made in good time and be acknowledged in writing by the relevant utility company, with confirmation that the services have or will be isolated or disconnected.

Prior to their arrival on site, utility companies should be sent any relevant sections of the pre-construction information to incorporate within their own safe systems of work (for example, details of ground contamination and/or unsafe conditions or structures prevailing on site).

Where such disconnection is not possible, any pipes or cables should be clearly identified, marked and protected to ensure that they are not disturbed during the works. Disconnection confirmation in writing must be received and be on site before any demolition work commences.

Finally, if in doubt over any services, the contractor should seek further advice from the utility provider, specialist services contractor or services consultants. If overhead power lines are present, care needs to be taken, particularly where lifting equipment, excavators or demolition machines with high reach are to be operated.

Adequate control measures should be put in place (such as warning goalposts) following advice from the relevant power supply company.



For further guidance refer to the NFDC document *Disconnection of services for demolition and refurbishment work places*.



For further information refer to Chapter D07 Underground and overhead services.

3.7.3 Lead

If proper precautions are not followed, there is a risk of exposure to lead paint through dust and fumes.

The risk of lead exposure is considered to be greater when working in pre-1970 buildings, especially during repair, maintenance, refurbishment and demolition.

It should be noted that, wherever paintwork/coatings are likely to be damaged or disturbed, the HSE advises lead (paint) surveys **regardless of the age of the building or the extent of the works**. Where lead exposure is likely to be significant an initial air monitoring survey to check how much lead is in the air may be required.



Information on reducing the health risks of lead construction work can be found on the HSE website.



For further information refer to Chapter B08 Lead.

3.7.4 Asbestos

Under Asbestos Regulations, there is a duty on building owners and/or the tenants to identify (via asbestos surveys) and manage the presence of asbestos in the non-domestic properties under their control. HSE publication HSG264 provides the following guidance.

- The types of survey.
- The circumstances in which either type of survey would be required.
- Appointing a competent asbestos surveyor.
- The exchange of information that is necessary between all interested parties.
- Avoiding caveats whereby some areas would not be surveyed.
- What should be expected by way of a survey report.

The regulations state that all asbestos-containing materials should be removed prior to demolition, so far as is reasonably practicable. They also require that the necessary planning actions and notifications are carried out.

The management process should be based on the information gained from a refurbishment or demolition asbestos survey, carried out before contractors are invited to tender for the demolition. This should include a drawing of the building footprint (all floors) and a list of the approximate amounts and locations of asbestos-containing materials found.

Simply writing 'throughout the building' is not acceptable and the client should be informed that the survey is inadequate. The survey should be checked for any exclusions (caveats) or areas not accessed and for any specific recommendations. However, it should be noted that such clauses are generally unacceptable, as detailed in HSG264.

If the survey identifies licensable asbestos materials (typically asbestos insulation, coatings and insulating board), which must be removed by a licensed contractor, then 14 days must be allowed for the notification process to the enforcing authority, usually the HSE. The licensed contractor will submit the notification online using the FODASB5 notification form.

All non-licensed work (for example, removal of asbestos cement sheets) needs to be carried out with the appropriate controls in place. However, for notifiable non-licensed work (NNLW) additional requirements have to be met, including notifying the relevant authority of the intention to work with or remove asbestos, ensuring medical examinations are carried out and maintaining registers of work. Some examples of NNLW are removal involving textured coatings, where the method of removal requires deterioration of the material, and removal of asbestos paper and cardboard products, if not firmly bonded in products.



Asbestos pipe lagging



Notification of notifiable non-licensed work can be made online using form ASB NNLW 1.



The client should ensure that appropriate surveys have been carried out and that the results are acted on accordingly.

From the survey, an inventory of asbestos-containing materials should be made and ticked off as they are removed. This should prevent creating risk during soft strip and demolition.

There are many occasions when asbestos-containing materials are only revealed during the demolition.

All operatives and machine drivers must receive mandatory annual asbestos awareness training so that they can recognise materials that could be, or contain, asbestos and stop work and take appropriate advice. This training does not prepare workers or self-employed contractors to carry out work with asbestos-containing materials. Training can be classroom based or offered as an e-learning course, provided it satisfies the requirements of Regulation 10 of the Control of Asbestos Regulations and the supporting Approved Code of Practice.

Workers who carry out work that will disturb asbestos (non-licensed work, notifiable non-licensed work and licensed work) must receive a higher level of training.



Company fined for employee asbestos exposure

A construction company has been fined for safety breaches after employees were exposed to asbestos after removing false ceiling tiles during a shop conversion in Hull.

Beverley Magistrates' Court heard that the company had not commissioned a refurbishment asbestos survey prior to the work commencing. Employees removed over 1,000 m² of asbestos insulation board (AIB) ceiling tiles in an uncontrolled manner, exposing them to asbestos.

An investigation by the HSE found that the company's director, and the casual labourers they employed, spent approximately three to four weeks removing the suspended ceiling, along with the ceiling tiles which contained asbestos, to install new stud walls to divide the shop floor into separate units.

The labourers were unskilled and untrained. They were provided with a claw hammer to knock the tiles down. The asbestos-containing tile debris was then shovelled or collected into approximately 62 one-tonne bags.

The company from Bradford, pleaded guilty to breaching Regulation 5 of the Control of Asbestos Regulations 2012. The company has been fined £16,000, ordered to pay £3,011.87 in costs and a victim surcharge of £190.

After the hearing, a HSE inspector commented: "If the company had identified any asbestos on the site through a refurbishment asbestos survey, carried out by a competent surveyor, and had it removed by licensed asbestos removal contractors prior to the refurbishment work commencing, then employees would not have been exposed to asbestos.

"No matter how small or large your company, there is a need to prevent exposing your employees and the public to asbestos by ensuring that it is identified on site prior to any work commencing."

(Source: HSE.)

For advice on suitable awareness training visit the HSE website.



Asbestos guidance note *Industry guidance for non-licensed work with asbestos containing materials within a demolition or refurbishment environment* is available from the NFDC.



For further information refer to Chapter B09 Asbestos.

3.8 Methods of demolition

This section gives an outline of the types of demolition techniques commonly employed, with a view to giving the reader an appreciation of the demolition activity.

The key to a successful demolition project is to ensure that the appropriate risk assessments and method statements have been developed to identify the correct sequence of carrying out the work, together with the necessary tools, equipment and people required.

These must be recorded in writing, fit for purpose, site and task specific, and clearly communicated and understood by the persons using them.

3.8.1 Partial demolition

Partial demolition is often carried out where refurbishment is being undertaken and can include façade retention. In any demolition, daily or, where required, more frequent checks should be carried out to confirm the stability of the remaining structure.

If at any time during the demolition the structure appears or becomes unsafe, all workers should be withdrawn until actions have been taken to remove any danger. This may involve the use of mini machines (such as skid steers and/or mini excavators).

Therefore, an assessment of the floor-loading capacity should be undertaken by a competent person (such as a structural engineer) to take into account the dead and live loads, and any necessity for temporary works (such as propping, shoring and so on).

3.8.2 Progressive (top-down) demolition

Progressive demolition is generally carried out in the reverse order to construction, and often follows the soft strip-out phase. This is the most commonly used method of demolishing structures and should be detailed in the construction phase plan.



Demolition of high-rise building

In high-rise buildings where a floor-by-floor demolition is being carried out, the following danger points should be recognised and adequate precautions taken.

- Structural stability.
- On-floor loadings (machinery and arisings).
- Falling debris.
- Slips, trips and falls.
- Maintaining clear access and egress.
- Risk of fire.
- Open and unprotected edges, including shafts and voids.



The NFDC has produced guidance on demolishing multistorey structures of 18 m and above, and in the use of high and ultra high-reach demolition machines, to improve procurer and user awareness, entitled *Guidance for deconstruction of tower blocks floor by floor/piecemeal*.

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3.8.3 Demolition by designed collapse mechanism

Demolition by designed collapse can be achieved by pre-weakening the structure, followed by explosive displacement of important structural elements, remote mechanical demolition by a demolition machine, high-reach demolition rig or crawler crane or similar, or occasionally by pulling it down using a wire rope.

Explosive displacement requires a high level of expertise, planning, supervision and execution. Consequently only qualified explosive demolition engineers should be involved in this type of demolition (for example, members of the Institute of Explosives Engineers).

3.8.4 Manual demolition techniques

Manual demolition techniques are used when other methods of demolition are not suitable or possible (for example, city centre or confined sites, where there is insufficient space around the structure for mechanical demolition).

Some of the types of tools or operations that can be used in manual demolition are listed below.

- Hand tools.
- Breakers, compressors or hammers.
- Concrete nibblers or hydraulic pulverisers.
- Stitch drilling.
- Drilling and hydraulic bursting.
- Drilling and expansive pastes.
- Oxy-propane cutting equipment.
- Diamond cutting and sawing.

For brick or concrete structures:

- wherever practicable, carry out demolition in the reverse order to construction
- identify any pre-stressed or post-tensioned concrete beams, columns and trusses that may be present within the structure and determine a safe method of demolition
- use compressed air, battery-powered tools or those from a 110 volt supply
- maintain tools in good condition, and use them safely
- make operatives fully aware of the safe procedures
- recognise the potential health risks from exposure to high levels of noise and vibration requiring assessments and the associated control measures
- in addition to site induction, ensure task and toolbox talks are prepared, delivered and understood at key stages of the work.

As far as is reasonably practicable, employees should not work above each other and care must be taken to ensure that debris does not drop into working or otherwise occupied areas.

If lift or service shafts or other openings through floors are used for the removal of debris, the lift car, services and equipment must be cleared in advance to remove any snagging points. Openings must be adequately protected by either suitable guard-rails and toe-boards (with no gap between guard-rails and toe-boards exceeding 470 mm) or by other substantial, effective barriers.

It may not be possible to guard an opening where plant (such as a skid steer) is being used to bulldoze arisings into a shaft or chute. In such cases a safe system of work must be developed, which is sufficiently robust to:

- protect other workers from falling down the shaft
- stop the item of plant falling into the chute or shaft
- control when materials will be loaded and unloaded to stop materials being tipped onto someone below.

DEMOLITION

It is still acceptable to use window openings as a means of removing debris from upper floors under certain conditions. The opening would have to be protected to prevent operatives falling whilst throwing the debris out. Furthermore, the landing zone (that is, the area to receive the materials arising from the soft stripping or demolition operations) must be clearly identified and completely protected so that materials cannot fall on anyone. Typically the area (drop zone) will be fenced off with mobile fence panels and then the material loaded with an excavator into a hook bin container; this area should be controlled by a vehicle marshaller to ensure that no entry can be effected by unauthorised persons.

If the structure has more than two storeys, consideration should be given to the prevailing wind conditions, use of an enclosed scaffold chute and fitting the receiving skip with suitable dust covers or shrouds. With regard to falling materials and exposed edges, where necessary, hazard areas (personnel exclusion zones) must be created in accordance with the requirements of current regulations and good practice.

3.8.4.1 Oxy-propane cutting equipment

The use of oxy-propane cutting equipment requires the following.

- Operatives are trained and competent in the safe use of the equipment.
- Appropriate PPE is used (such as the correct respiratory protective equipment (RPE), goggles and hand or arm protection).
- The equipment is inspected and tested for leaks before use.
- Cylinders are secured in an upright position, at least 3 m from any party wall, boundary fence line or hoarding.
- Hoses are secured with crimped fittings, not jubilee clips.
- Flashback arresters are always fitted between cylinder gauges and hoses.
- Spare cylinder storage areas are recommended to be separated by a gap of at least 3 m between oxygen and propane bottles.
- A hot-work permit system is in place and is followed.

Do **not**:

- operate hot cutting equipment within an enclosed space, because of fumes, leakage of fuel and/or oxygen gases
- use oxygen or compressed air to blow dust off clothing
- carry or use petrol or gas lighters to ignite the cutting torch (use only approved flint guns)
- rely on existing site fire points (ensure additional fire extinguishers are in place prior to commencement of hot works).

Whenever oxy-propane cutting equipment is used, the correct fire-fighting equipment must always be available. All operatives involved in the work should be trained in the safe use of fire extinguishers.

Hot work should generally be stopped one or two hours before the site closes and be regularly monitored to avoid the potential of smouldering material developing into a fire unobserved.



For further information refer to Chapter C02 Fire prevention and control.

3.8.5 Mechanical demolition techniques

Machinery should be fit for purpose (in line with PUWER), operated by authorised, competent persons (for example, those holding a Construction Plant Competence Scheme (CPCS) card for the appropriate category of machine) and be used in safe working spaces.

Because of the danger of debris falling onto the demolition rig and its driver, the machine should be a safe distance from the structure and be fitted with roll-over protective structures (ROPS) and/or falling object protective structures (FOPS). As an added precaution, it should be fitted with shatterproof glass.

Before using the demolition equipment, steps must be taken to ensure that the building is completely empty and that all services are isolated with the appropriate written confirmation of disconnection on site.

3.8.5.1 Remote control demolition

Remote demolition is generally carried out by specialised equipment, often in hazardous or aggressive environments (for example, nuclear installations). The use of pedestrian operated machines is becoming more common and they offer possible solutions to the difficulties of complying with the problems of noise and vibration experienced during hand demolition.

These types of machine enable the operator to stand well clear of danger areas. They are particularly useful where the machine is breaking out the structure of the floor above, and/or when the operating surface is being progressively removed.



ROPS and FOPS cab

3.8.5.2 Using a 360° adapted demolition rig with multi-functional attachments

This type of demolition machine is commonly used to demolish relatively low-level buildings, or is used after other height reduction techniques have been carried out.

The machine should always be positioned at a safe distance from the structure, taking into account the height of the structure and the build up of debris as it is reduced. In situations where the machine operator cannot see all elevations of the structure or in circumstances where external influences may dictate stopping the work until such distractions have been removed, a vehicle marshaller or lookout may be required to assist the operator. The vehicle marshaller or lookout should be positioned in a safe location, outside of the machine's operational exclusion zone, and preferably within sight of the operator (or in radio contact, where this is not possible).

Whereas the machine driver is isolated from noise and dust, the vehicle marshaller or lookout is not and may need to wear additional PPE, depending on the findings of the risk assessments for the work being done (for example, impact-resistant goggles if there is a possibility of debris being thrown off).

The height of the wall or building to be demolished should not normally be greater than the maximum reach of the machine; this maintains a safe distance between the machine and the structure being demolished.

In some circumstances, it is appropriate and practicable to create a ramp for the machine to sit on to increase the reach using rubble from previous demolition. Care must be taken to ensure the ramp and operating platform are properly designed and constructed and that there has been sufficient compaction to avoid the machine sitting on an unstable base.

It is not good practice to construct a ramp within a building, due to potentially overloading the floors or surcharging the outside walls.

It is not recommended to allow undermining or undercutting of structures when the machine cannot reach the top of the building. Any contractor suggesting this method should consider selecting a larger (higher reach) machine to avoid putting the operator at unnecessary risk from falling structures and debris.

3.8.5.3 Super high-reach 360° excavators

Typically these machines have a reach of between 15 and 65 m (or more), plus the length of the fitted attachments for shearing concrete or steel structural elements. To ensure a safe distance between the machine and the structure, a ratio of 2:1 height to distance must be maintained for machines up to 45 m reach. In some cases this may mean that the recommended maximum working height for the machine is restricted to 75% of its maximum reach. A useful visual indicator that the machine is working within safe limits is when the dipper (the last section of the excavator's arm) is horizontal or approximately 90° to the boom.

These sophisticated machines are fitted with variable width tracks and are heavier than standard machines, thus giving increased ground-bearing pressure and extended working envelopes. It is vital that ground conditions are known and any voids and ducts, located where the machine will track, are identified, clearly marked (fenced off) and, where possible, adequately filled and compacted. In terms of ground conditions, these machines should be considered more as a crane or piling frame in terms of stability, rather than an excavator.

Operator training is supported by a specialist CPCS category for high-reach machines with various endorsements for different reach machines (for example, up to 15 m, 30 m and 30 m+).

It is essential for operators to understand how to operate the machines safely and also have knowledge of good practice with regard to fitting and changing attachments. The manufacturer or the supplier of the machinery can provide advice on this. There are further CPCS card endorsements for this activity.



Ground conditions should be assessed before using super high-reach excavators



High reach demolition rig guidance notes can be downloaded free from the National Federation of Demolition Contractors.

3.8.5.4 Hydraulic pusher arm

Demolition using a hydraulic pusher arm is occasionally used for removing masonry structures. However, with the advent of super high-reach machines with multi-functional attachments, it is almost obsolete now and its use should only be carried out by experts. It is therefore not covered within the scope of this chapter.

3.8.5.5 Demolition ball

Demolition using a ball is extremely rare due to the advent of super high-reach machines. When a ball is used, the lifting equipment must be heavy duty and only drop or pendulum (for example, in line with the jib) balling techniques should be employed. When a ball is employed, regular (for example, hourly) inspections of the equipment must take place, paying particular attention to the attachments and shackles. Arrangements for safely rectifying the situation should be in place in the event of the demolition ball becoming trapped within the structure. Further detail is not appropriate within the scope of this chapter.

3.8.6 Bridges or steel structure demolition

The demolition of bridges, pylons, masts, or similar, requires specialised planning and techniques. The information provided below can be used as a guide, but specific detail is not covered within the scope of this chapter. An assessment by a demolition engineer and/or a structural engineer should be undertaken. A comprehensive safe system of work must be developed providing a safe means of access, using a competent crane hire operator and experienced slingers and signallers to ensure compliance with the Lifting Operations and Lifting Equipment Regulations 1998. For bridges that run over roads or cross railway lines, liaison must take place with the Highways Department of the Local Authority involved or Network Rail, as appropriate.



For further information refer to Chapter C07 Lifting operations.

3.9 Environmental and other considerations

Noise, dust, fumes, vibration and fire controls need to be properly addressed before and during the demolition operation. Consideration should be given to the following points.

3.9.1 Noise

Where demolition operations are carried out in locations likely to affect the general public, the Local Authority may request the principal contractor to submit an application under Section 61 of the Control of Pollution Act 1974, specifying working methods, working hours and maximum noise levels (called prior consent). A Local Authority can issue an abatement notice (Section 60 notice) for failure to comply with Section 61 consent conditions, or where noise levels are creating a statutory nuisance. Checks should also be made for any constraints in site working hours (such as planning permission conditions or the Local Authority or council's own restrictions on working hours).

In accordance with regulations, where noise levels are above (or expected to be above) the lower exposure action value, the demolition contractor must ensure that a noise assessment has been carried out and that, where possible, people are kept out of the danger area.



A monitor to assess periodic noise

Any machinery that is to be used in the demolition process should, as far as possible, be fitted and used with soundproofing or noise reducing equipment (such as exhaust silencers). Where it is necessary for people to work within the area of noisy operations, adequate hearing protection must be provided and used as necessary. If the upper exposure action value, 85 dB(A), is exceeded, or is likely to be exceeded, hearing protection must be worn and hearing protection zones clearly indicated with mandatory signs.

Where the findings of a risk assessment indicate that the hearing of any employee is at risk due to noise exposure at work, health surveillance, including hearing checks, must be provided.



BS 5228 gives advice on the provisions for noise and vibration control on demolition sites. For further guidance refer to Chapter E06 Statutory nuisance.

3.9.2 Dust

Nearly all demolition work creates dust and requires a control of substances hazardous to health (COSHH) assessment (for example, respirable crystalline silica). Taking simple precautions (such as those listed below) can help to reduce dust generation and exposure.

- The use of light water sprays both before and during demolition are effective. However, consideration should be given to any potential run-off contaminants that may be produced and to the proximity of demolition work to electrical services, retained structures and drains.
- Where demolition is being carried out inside a building and water sprays are inappropriate, local ventilation, using air movers and filters, can help to alleviate dust levels (it may also be possible to create voids in the structure by taking out, for instance, walls and partitions).
- Consideration should be given to identifying the type of dust the operative is exposed to and measuring the actual dust levels to confirm whether or not the anticipated levels are being achieved and thereby whether the measures being taken (for example, personal and perimeter pumps) are effective.
- Respiratory protective equipment, as with any other PPE, should be used only as a last resort. Most forms of respiratory protective equipment require the wearer to be **face-fit tested** before use.



For further guidance refer to Chapter B10 Dust and fumes (Respiratory hazards) and Chapter E06 Statutory nuisance.

3.9.3 Fumes and gases

A cutting torch, used on steelwork, may produce toxic gases (such as nitrogen dioxide). If a phosphate coating is present, phosphine may be produced. If a chlorinated solvent has been used, sulphides may be formed, which have no smell until high toxic levels are present. Toxic metal fumes may additionally be given off.

Attention is drawn particularly to lead (lead-painted steelwork), cadmium (cadmium bolt heads) and zinc.

Before any hot work cutting is allowed, available information or paint samples may be required for analysis, a COSHH assessment must be undertaken, and the necessary control measures implemented.

Operatives undertaking hot work must be issued with the appropriate PPE, including appropriate respiratory protective equipment (RPE), as necessary.

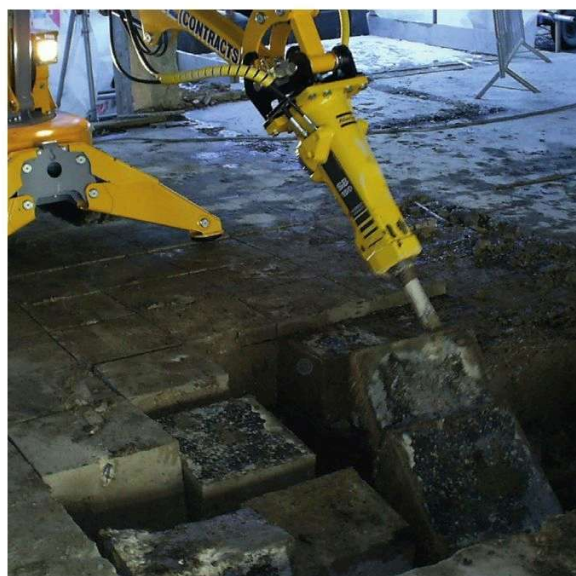


For further information refer to Chapter B07 Control of substances hazardous to health, Chapter B08 Lead, Chapter B10 Dust and fumes (Respiratory hazards) and Chapter E06 Statutory nuisance.

3.9.4 Vibration

Vibration from demolition operations can cause damage to adjacent property and injury to personnel working on site or to members of the public.

Exposure to vibration must be minimised and controlled in accordance with the requirements of the Control of Vibration at Work Regulations 2005.



Using a machine-mounted breaker instead of a hand-held version will eliminate the risks associated with hand-arm vibration

- Where possible avoid exposing people and/or property to vibration by selecting appropriate demolition techniques.
- Attempts should be made to establish the presence of any existing sources of vibration and whether vibration monitors are needed in sensitive locations.
- Where buildings adjoining those to be demolished are being retained, separation should be carried out using hand tools rather than machinery.
- Necessary precautions should be taken or alternative equipment considered to alleviate the risk of permanent physical injury, such as hand-arm vibration syndrome (HAVS) (for example, vibration white finger, from continued use of vibrating tools) or whole-body vibration (WBV) from continued exposure to vibration from mobile plant operation.

With regard to the last point, the exposure time for most demolition hand tools is extremely short, and the contractor will need a robust policy that includes health surveillance to carry out hand demolition using pneumatic electric or hydraulic demolition picks.



For further information refer to Chapter B12 Vibration and Chapter E06 Statutory nuisance.



For further information refer to the Code of Practice for noise and vibration control on construction and open sites (BS 5228).

3.9.5 Fire and explosion risks

Where flammable liquids, gases or vapours have been used, or released in a building that is under demolition, any equipment, tank or pipes, which could have contained such substances, must be purged and tested for explosive gases prior to work taking place. Any such work should be done under a permit-to-work system.

General hot work, using oxygen and propane cutting equipment, should be carried out only by operatives trained in its safe and proper use, wearing the appropriate PPE, including goggles, gloves, overalls and respiratory protective equipment.

Before any burning of debris is undertaken, consideration needs to be given to waste management licensing, or any local byelaws, which may prohibit such actions taking place. Penalties for the contravention of such conditions can be severe in some circumstances.

If the occasional burning of debris (such as wood or paper) is allowed on site, the fire must be as small as possible, well away from buildings, roadways and fuel stores, and kept under constant supervision.

All fires must be completely extinguished at least one hour before work stops for the day and checked again, to ensure there are no glowing embers, before operatives leave the site. Adequate fire-fighting equipment must be available, with fire points containing extinguishers in prominent and well-marked areas adjacent to the demolition hot-work operations.



For further information refer to Chapter C02 Fire prevention and control.

3.9.6 Recycling

Demolition works should always consider opportunities for reuse, reclamation and/or recycling of materials. A material survey of the site will identify what materials are on site and how they may be recovered, managed and removed to maximise recovery. Where relevant, the contractor can opt to use a site waste management plan to record such material.



For further details refer to Chapter E10 Waste and material management.

In addition to removing recoverable items from demolition operations, it is common to crush the resultant brick and concrete into a sub-base (hard core) material for future construction purposes.

- Crushing on site will generally take place when environmental conditions permit such actions. Environmental emissions (noise, fumes and dust) are considered as statutory nuisances and should therefore be eliminated or reduced to the lowest possible level. Regular monitoring of all control measures are required daily and should be recorded on a check sheet. Operations should cease if there are excessive emissions, which lead to nuisance being caused to people outside of the site boundary.
- Mobile crushers must be registered with the supplier's own Local Authority, which must be notified in advance each time the equipment is moved.
- All materials to be recycled should be checked for contaminants and dealt with in the correct manner to the required specification. The reuse of construction and demolition waste may require an environmental permit or relevant exemption. Advice should be sought from the local environment agency waste licensing officer on a case-by-case basis where any doubt may arise.
- Operatives working on the crushers and screeners must be properly trained (both general training on the use of crushers and screeners and specific training in respect of the equipment being used).
- Serious consideration must be given to the need for machinery guarding. A daily check sheet should be completed by the operator, which confirms that all guards are in place and the emergency stops are working.
- Robust safe systems of work are required to deal with blockages. These often happen because the operator loading the crusher or screener overfeeds the belt or because lumps are too big for the machine to process. The machine operator feeding the crusher or screener should sort the materials prior to loading by removing the timber, metals, plastics and fabrics that have entered the stockpile.
- All crusher and screener operators are required to wear suitable and adequate hearing protection and may require hearing checks and monitoring as the noise levels can be high.
- Dust suppression systems can be an effective, operational, engineering control, if fitted and working correctly. Reusable half mask respirators are preferred to the use of disposable paper masks as these will offer a higher level of protection. A risk assessment will be required to determine the appropriate type of RPE. Face-fit testing will be required following this.
- Operators should not be allowed to stand on an operating crusher, and must never stand near the jaws where the feedstock is being crushed. The risks arising from this practice can be combatted at source by effective soft stripping and, if necessary, allowing picking from a product feed belt (in other words, before or after it has gone through the crusher).

3.9.7 Removal and disposal of materials

Before any materials are removed from site, both vehicular access to, and egress from, the site must be agreed with all concerned.

It is prudent and practicable to leave the floor slabs, roads and hard standings in place until the end of the project to avoid mud and other debris being carried out onto the highway. Where necessary, bog mats, to create mud free stable access routes, and wheel-washing facilities should be provided to ensure that mud and debris are not carried onto the highway.